

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- Midterms are graded; returned through Gradescope this afternoon.
- Midterm revisions due next Friday.
- This week: Chapter 4
- Quiz 8 today. (Computing determinants.); HW6 distributed today.
- Problem session Friday, 3-5pm in Ryerson 177.
- Final three homeworks (written and WeBWorK) due on Mondays at 11:59pm. (Three days later than previously.)
- Sunday HW5 Debrief (“Prebrief”) Cobb 115, 3-4pm
- Monday requizzing session, Quizzes 1-8, 3:30-4:30pm in Eckhart 207A
- Problem swap 2 problems due next Wednesday.

Midterm exam (2021)

class average

76

Midterm exam (2021)

class average

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Midterm exam (2023)

class average

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Midterm exam

Midterm exam

Revisions: by reworking the questions you missed on the exam, you can earn back 1 point for every 5 that you missed.

80 → 84

Due next Friday at 11:59pm on Canvas.

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- improve on the final
- do more flex
- hope for a curve

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63	70	80	B-
50	60	76	C

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81	85	91	A-
63	70	82	B-
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class average & median	81	85	91	A-
	→ 76	81	88	B+
	63	70	82	B-
	50	60	76	C

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	81	85	91	A-
no flex	100	100	90	A-
max flex	73	78	86	A-

Ch 1-3: fields have properties *in common*.

Ch 4+: fields have properties *that are different*.

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Ch 4+: fields have properties *that are different*.

Does $\mathbb{F}$ have a consistent notion of “size”?

Is $\mathbb{F}$ algebraically complete?

Is $\mathbb{F}$ continuous or discrete?

(Does every polynomial in $\mathbb{F}[x]$ have a root?)

(Can elements of $\mathbb{F}$ be arbitrarily close?)

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$\mathbb{R}$

$\mathbb{C}$

$\mathbb{Z}/5\mathbb{Z}$



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$\overline{\mathbb{R}} = \mathbb{C}$ Theorem: every field has another field that's its algebraic closure. Proof: Zorn's Lemma.

Fundamental Theorem of Algebra

Every non-constant polynomial in $\mathbb{C}[x]$ has a root in $\mathbb{C}$.

Consequently, every non-constant polynomial in $\mathbb{C}[x]$ of degree n has exactly n roots in $\mathbb{C}$, counting with multiplicity.

algebraic multiplicity

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algebraic multiplicity

Every operator on $\mathbb{C}^n$ has exactly n eigenvalues in $\mathbb{C}$, counted with multiplicity.

On the other hand, an operator on $\mathbb{R}^n$ may have *no* eigenvalues in $\mathbb{R}$.
(And similarly for $(\mathbb{Z}/5\mathbb{Z})^n$.)

Find the eigenvalues and eigenvectors of $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

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$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix}$$

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$$\det(A - \lambda I) = \lambda^2 + 1$$

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Find null $(A - iI)$: $\begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix}$

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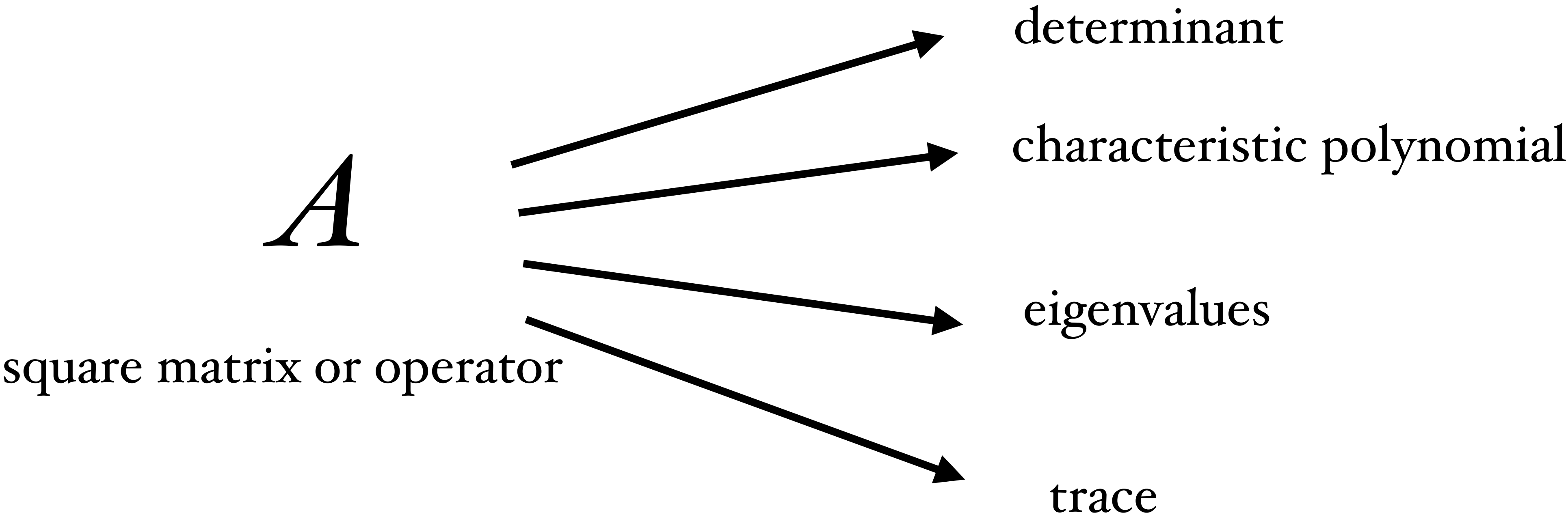
$$\det(A - \lambda I) = \lambda^2 + 1$$

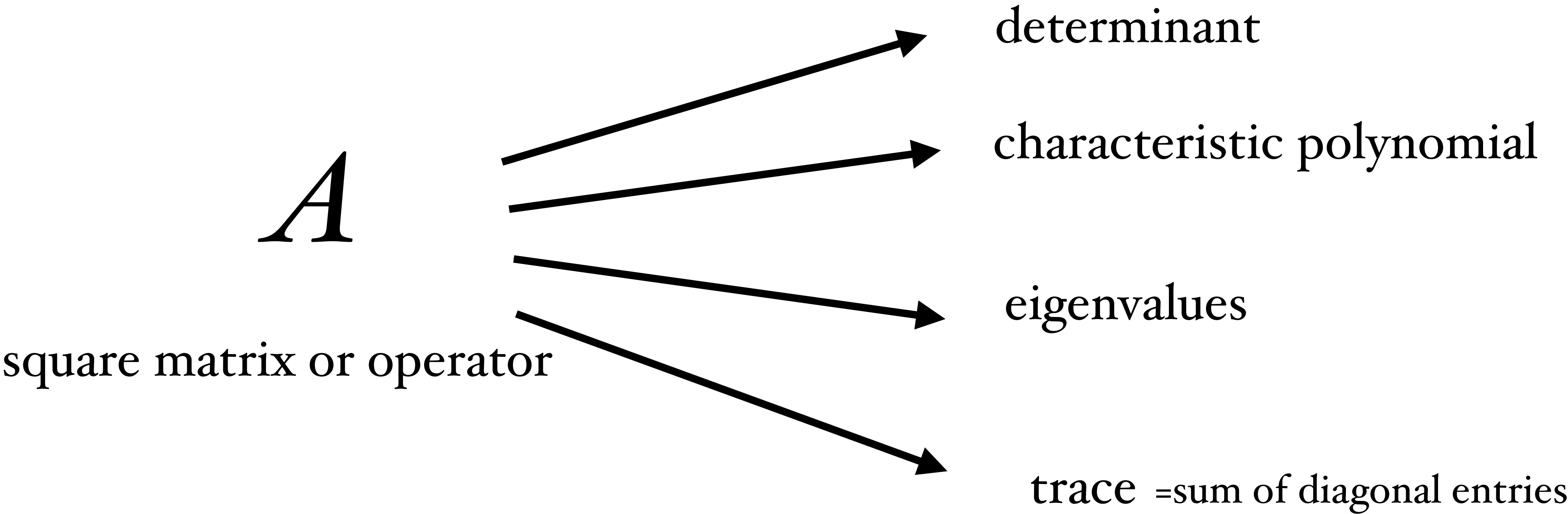
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Similarly, $\begin{pmatrix} 1 \\ i \end{pmatrix}$ is a basis for the $-i$ -eigenspace of A .





These are invariants of
the underlying linear transformation
and so are invariant under change of basis.

A

square matrix or operator

determinant

characteristic polynomial

eigenvalues

trace =sum of diagonal entries

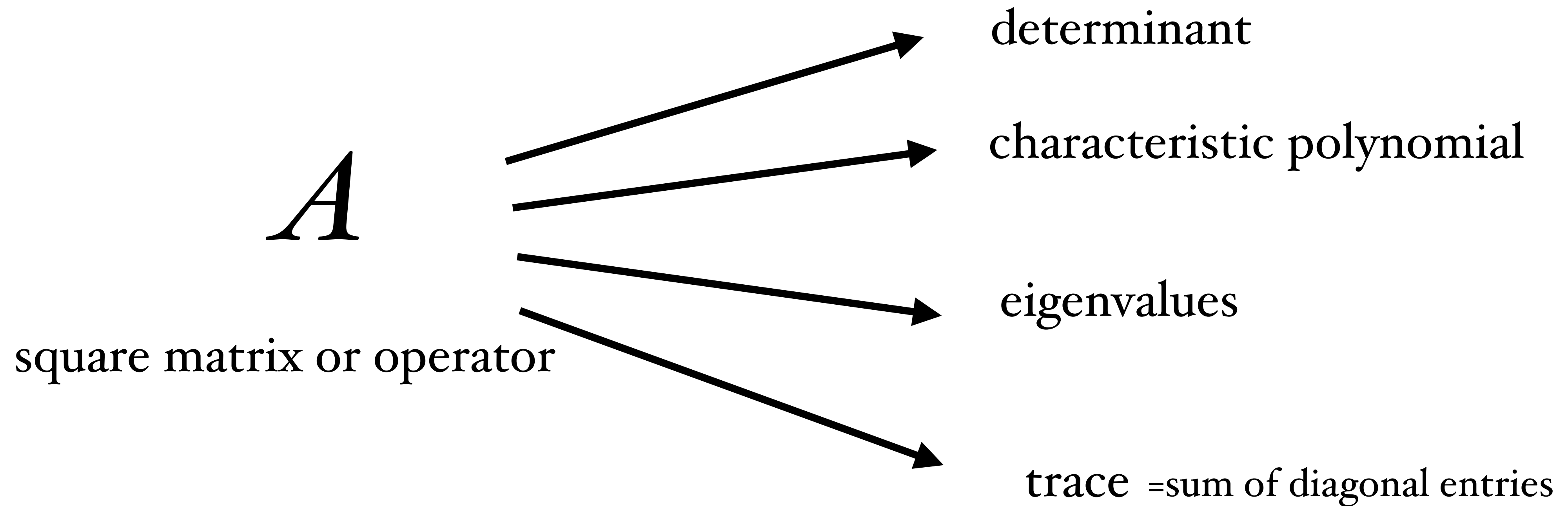
Object

Invariants

(shadows/properties)

1. How “good” are the invariants?
2. How do the invariants interrelate?

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Object

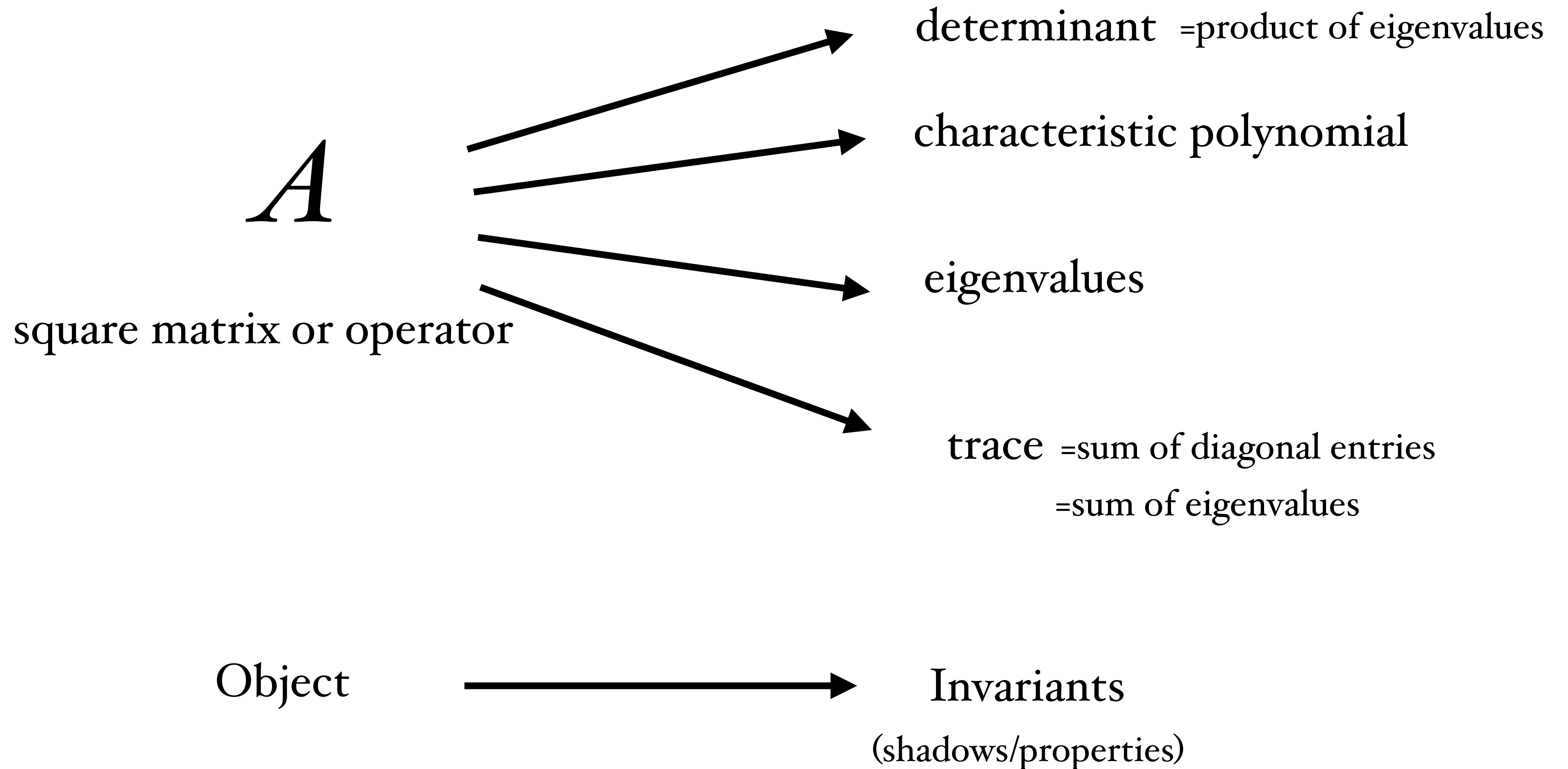


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$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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determinant:

trace:

characteristic
polynomial:

eigenvalues
(with multiplicity):

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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determinant:

I

I

trace:

characteristic
polynomial:

eigenvalues
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$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

determinant: **I**

I

trace: **2**

2

characteristic
polynomial:

eigenvalues
(with multiplicity):

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

determinant:

$$1$$

$$1$$

trace:

$$2$$

$$2$$

characteristic
polynomial:

$$(\lambda - 1)(\lambda - 1) = \lambda^2 - 2\lambda + 1$$

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eigenvalues
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eigenspaces:

$$x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \dim 2$$

$$x \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \dim 1$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

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geometric multiplicity

$$x \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \dim 1$$

WU

For eigenvectors and eigenvalues, diagonal matrices are nice!

For a given operator, is there a basis where the corresponding matrix is diagonal?

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1. Find the eigenvalues and eigenvectors of A .
2. If geometric multiplicity = algebraic multiplicity, proceed! Else, can't be diagonalized.
3. Use “change of basis” matrices to write $A = QDQ^{-1}$, where D has the eigenvalues of A on the diagonal and Q has the corresponding eigenvectors as its columns.

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$$\lambda_1 = 1, v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$$
$$\lambda_2 = 2, v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

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